

# Pramo Jewels

## AI API Development Guide

Document ID: PJ-AI-API-001  
Version: 1.0  
Status: Approved

| Field    | Value                                                                         |
|----------|-------------------------------------------------------------------------------|
| Project  | Pramo Jewels Premium E-Commerce Platform                                      |
| Purpose  | Define standards for AI-generated API development, security and documentation |
| Audience | Backend Developers, API Engineers, AI Coding Assistants, QA                   |
| Version  | 1.0                                                                           |

### AI API Development Standards

| Category       | Guidelines                                                 |
|----------------|------------------------------------------------------------|
| API Design     | RESTful endpoints, versioning and consistent URI structure |
| Validation     | Request validation and standard error handling             |
| Authentication | JWT/session auth and RBAC                                  |
| Documentation  | OpenAPI/Swagger with examples                              |
| Performance    | Pagination, caching and rate limiting                      |
| Security       | HTTPS, sanitization and secret protection                  |
| Observability  | Logging, tracing and metrics                               |
| Testing        | Unit, integration, contract and load testing               |

### Objectives

Ensure AI-generated APIs are secure, scalable and aligned with project architecture.

### Design Rules

Use semantic HTTP methods, resource-oriented endpoints, idempotency where applicable and standard status codes.

### Response Standards

Return consistent JSON payloads with metadata, pagination and standardized error objects.

### Security

Enforce authentication, authorization, input validation, rate limiting and never expose secrets.

## **Performance**

Optimize payloads, database queries, caching and asynchronous processing.

## **Documentation**

Maintain OpenAPI specifications, examples and API version history.

## **Integration**

Integrates with Authentication, Database, CMS, Payments, Shipping, Notifications, Monitoring and CI/CD.

## **Acceptance Criteria**

APIs pass functional, security, performance and contract testing and conform to project standards.